Anand Bhalerao

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Experience

FIS Global **Pune, India**

Software Engineer 2 *Jan 2026 - Present*

- Shipped **real-time session revocation** for 40+ banking clients after a **$2M fraud incident**, using atomic invalidation (Spring + PostgreSQL) to kill compromised sessions across multi-tenant systems with zero downtime.
- Architected **field-level AES encryption** as a custom MyBatis interceptor that rewrites queries transparently — securing **1M+ PII records**, adopted by 6 engineering teams.
- Engineered a resumable, chunked AES **key-rotation pipeline** with pre-flight validation, version tracking, and **idempotent re-encryption** - rotating **1M+ records** across 10+ environments with full rollback and zero unsafe rotations.
- Cut encrypted-field lookups from **47s to 60ms via blind indexing** with clustered indexes, without weakening field-level encryption.
- Led **lazy credential-hardening** migrations at login: **500K+** password and challenge-answer hashes from SHA-256 to BCrypt; migrated AES MFA tokens and Base64-encoded OOBA tokens to PBKDF2 hashes, with **algorithm tagging** for future upgrades without bulk reprocessing or user disruption.
- Resolved an IBM MQ **transaction race** causing consumers to process messages before DB commits, leaving records in inconsistent states; implemented failure-state handling for reliable recovery.

Software Engineer 1 *Jul 2024 - Jan 2026*

- Eliminated **burst-traffic** request timeouts by replacing a single-row sequence table with block-based cached allocation, **removing row-lock contention** under load.
- Pushed pagination, sorting, and filtering into the database, cutting rows fetched per request from **90K to 10** on 100K+ datasets.
- Led **end-to-end Zelle SSO** for **40+** banking clients, reconciling external Zelle records against the bank's source of truth before SSO and automating account lifecycle workflows.
- Redesigned multi-entity session handling, ending recurring user lockouts and enabling account switching without re-authentication.

Software Development Engineer Intern *Jan 2024 - Mar 2024*

- Delivered Electronic File Delivery enrollment end to end - 8 Angular screens, 6 Spring REST APIs.
- Built an interstitial enforcing mandatory contact updates, with admin-controlled retry logic to drive compliance.

Projects

Distributed UPI Payment Simulator **Demo Video**

Sarvatra Technologies

- Architected a **distributed UPI simulator** modeled on NPCI switch architecture using Spring Boot, Redis, and Kafka.
- Made payment processing fault-tolerant with idempotency and **Saga-based distributed transactions**; Redis Bloom filters cut database lookups by 92%.
- Load tested to **1,800 TPS** at **85ms p95 / 140ms p99**, instrumented end to end with OpenTelemetry + Prometheus.
- Deployed multi-replica Docker workloads on **Kubernetes** (k3s) over AWS EC2, with GitHub Actions CI/CD for testing, image publishing, and zero-downtime rolling updates.
- Secured access with JWT session management and **FIDO2/WebAuthn** passwordless authentication.

Education

Pune Institute of Computer Technology **Jun 2020 - Jun 2024**

B.E in Computer Engineering (CGPA - 8.94) *Pune, Maharashtra*

Achievements

- **1100+ problems** solved across LeetCode (peak 1763), CodeChef (AIR 145, 496 in Starters), and Codeforces.
- Finalist in Smart India Hackathon '22.

Technical Skills

Languages: Java, Kotlin, Typescript
Application Development: Spring Boot, MyBatis, Hibernate, JPA, Angular, Compose Multiplatform
Databases & Cache: PostgreSQL, Oracle, MySQL, Redis
Infrastructure & Observability: Kafka, OpenTelemetry, Prometheus